From Centralized Remote Control to ROS 2 Onboard: A Retrofitting Architecture for a Legacy Soccer Robot

Leffer Trochez
Dept. of Electrical and Electronic Engineering
Universidad de los Andes
Bogota D.C., Colombia
l.trochez@uniandes.edu.co

Jorge Lopez-Jimenez
Dept. of Electrical and Electronic Engineering
Universidad de los Andes
Bogotá D.C., Colombia
jorgelopez@uniandes.edu.co

***Abstract*—Former robot soccer platforms were commonly designed under a remote-brainless architecture, where perception, decision-making, and control were executed on an external host computer, while the robot only received motion commands through wireless communication. Although effective for early FIRA competitions, this architecture limits onboard sensing, modularity, and compatibility with current robotic development workflows. This paper presents a retrofitting architecture for a legacy FIRA differential robot, taking it onto a ROS 2-enabled embedded platform. The proposed approach preserves the original mechanical structure while replacing the obsolete electronic and communication system with a modular architecture based on ESP32, ESP32-CAM, onboard sensors, motor drivers, Bluetooth/Wi-Fi communication, and micro-ROS integration. The resulting prototype demonstrates functionality related to onboard sensing, camera streaming, power monitoring, teleoperation, and expandable hardware interfaces through a ROS 2 node-based software structure. Rather than presenting a final autonomous robot, this work focuses on the architectural transition from a centralized legacy system to a modern, modular, and extensible platform for education, research, and future multi-robot development. The proposed architecture demonstrates a practical approach for modernizing obsolete robot soccer hardware to contemporary ROS 2-based robotic ecosystems.**

***Index Terms*—Legacy robots, robot retrofitting, robot soccer, FIRA, ROS 2, micro-ROS, embedded systems, differential-drive robots.**

I. Introduction

Robot soccer has historically served as a platform for developing mobile robotics technologies, including perception, communication, motion control, and multi-robot coordination. Early FIRA platforms commonly employed centralized, remote-brainless architectures in which an external host computer processed visual information, determined strategies, and transmitted motion commands to individual robots. Consequently, the robots operated primarily as mobile actuators, with sensing and decision-making performed externally.

Although effective in their original competition context, these architectures present significant challenges for contemporary robotics education and research. Their dependence on obsolete components, including overhead CCD cameras, frame grabbers, radio-frequency communication systems, and legacy software, limits their continued use. Nevertheless, their mechanical structures, motors, and differential-drive configurations often remain functional, providing an opportunity for modernization without complete mechanical redesign.

Contemporary robotics development increasingly relies on modular and distributed architectures that facilitate hardware integration, software reuse, and system extensibility. ROS 2 provides a node-based environment for integrating robotic components, while micro-ROS extends this approach to resource-constrained embedded devices. Although numerous educational and research platforms have adopted these technologies, modernizing existing robotic hardware introduces distinct challenges related to preserving mechanical components, replacing obsolete electronics, and integrating contemporary communication and software interfaces.

This paper addresses these challenges through a modular retrofitting architecture for a legacy FIRA differential-drive robot. The proposed approach preserves the original mechanical platform while replacing its obsolete electronic and communication subsystems with an embedded architecture incorporating onboard sensing, motor control, wireless communication, and standardized software interfaces. An ESP32-based controller manages sensing and actuation, while an ESP32-CAM provides onboard video streaming. Integration with ROS 2 and micro-ROS enables communication between the embedded platform and a host computer responsible for supervision, teleoperation, and software integration.

The principal contribution is a reusable architectural approach for transitioning mechanically functional legacy soccer robots from centralized remote control to modular embedded operation. Rather than introducing a new autonomous robotic platform or claiming quantitative performance improvements, this work focuses on the architectural organization, component integration, and implementation of a functional prototype. The resulting architecture provides a foundation for extending legacy robotic platforms toward contemporary educational and research applications.

The specific contributions are summarized as follows:

- A layered retrofitting architecture that preserves the me-

chanical structure of a legacy FIRA robot while replacing its obsolete electronic, communication, and software subsystems.
- A modular embedded implementation integrating onboard sensing, motor actuation, camera streaming, power monitoring, wireless communication, and hardware expansion interfaces.
- A ROS 2/micro-ROS software organization that supports bidirectional communication, teleoperation, sensor monitoring, actuator management, and future functional extensions.

The remainder of this paper is organized as follows. Section II presents the background and motivation. Section III describes the original platform and its limitations. Section IV introduces the proposed architecture, and Section V describes its prototype implementation. Section VI presents the architectural assessment, followed by the discussion in Section VII. Section VIII addresses limitations and future work. Finally, Section IX concludes the paper.

II. Background and Motivation

A. Legacy Robot Soccer Architectures

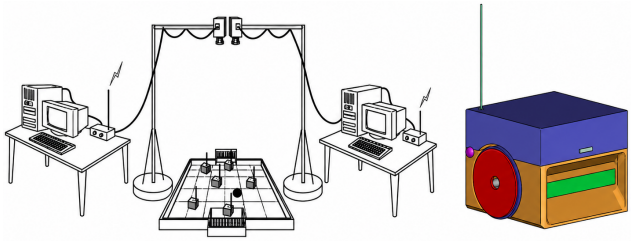
Robot soccer has been widely used as a benchmark for mobile robotics, multi-agent coordination, real-time perception, and control. Early FIRA/MiroSot and former RoboCup small-size platforms commonly relied on compact differential-drive robots operating in structured fields, where perception and decision-making were performed outside the robot. In these systems, an overhead camera observed the field, a host computer estimated the position of the robots and the ball, and wireless commands were transmitted to the robots for execution.

Several works describe this centralized vision-based paradigm. Kim et al. presented the CENDORI MIROSOT system, where robot control relied on the vision system and host-side generation of low-level motion commands [1]. Bizdoaca et al. described remote-brainless robot soccer systems, in which the host unit performs most of the computation while the robots execute transmitted commands [2]. Although effective in their original competition context, these architectures depend on external vision hardware, radio-frequency communication, and legacy host-side software, making many mechanically functional platforms difficult to reuse today.

B. ROS 2 and micro-ROS for Embedded Robotics

Modern robotic systems increasingly rely on modular and distributed software architectures. ROS 2 adopts DDS-based communication and provides a node-based computational model with publish/subscribe interfaces, configurable Quality of Service, and support for distributed execution [3]. These features are useful for replacing monolithic or host-dependent legacy control systems with modular robotic software components.

micro-ROS extends ROS 2 concepts to resource-constrained microcontrollers. In a typical architecture, a microcontroller runs a micro-ROS client, while a micro-ROS agent bridges the embedded device with the ROS 2 network. This communication is commonly supported through Micro XRCE-DDS, which is designed for constrained devices that exchange data with DDS-based systems [4], [5]. For legacy robot soccer platforms, this enables the robot to evolve from a passive command receiver into an embedded node capable of publishing sensor data, receiving actuator commands, and interacting with ROS 2 tools for teleoperation, monitoring, and future autonomous extensions.



(a) Legacy robot soccer setup (b) YSR-A robot platform

Fig. 1. Legacy FIRA robot soccer setup and the YSR-A robot platform.

C. Modular Educational Robot Platforms

Recent educational and research-oriented mobile robots emphasize low-cost, modular, expandable, and ROS-compatible designs. MAFOSS proposes an open-source multi-agent framework based on differential-drive mobile robots and ROS [6], ROMR presents a ROS-based open-source mobile robot with modular hardware and sensing capabilities [7], and SMARTmBOT demonstrates a low-cost ROS 2 mobile platform with onboard sensors, camera streaming, and educational applications [8]. These works demonstrate the benefits of accessible robotic platforms that combine embedded sensing, open software, and middleware compatibility.

Beyond newly developed platforms, some studies have explored the modernization of legacy robotic systems through retrofitting. Bomfim et al. preserved the mechanical structure of an industrial robot while replacing its obsolete control electronics with an open architecture [9]. More recently, Samson et al. upgraded a legacy automated guided vehicle using contemporary sensing and embedded technologies while maintaining the original platform [10]. However, these works address application-specific modernization rather than a reusable ROS 2/micro-ROS architecture for legacy robot soccer platforms.

This work addresses that gap by proposing a reusable ROS 2/micro-ROS retrofitting architecture that transforms a legacy FIRA robot from a remote-brainless platform into an embedded robotic system while preserving its original mechanical structure.

III. Legacy Platform

The case study considered in this work is a legacy FIRA differential-drive robot originally developed for small-size robot soccer competitions. The platform belongs to a generation of educational and competition-oriented robots in which

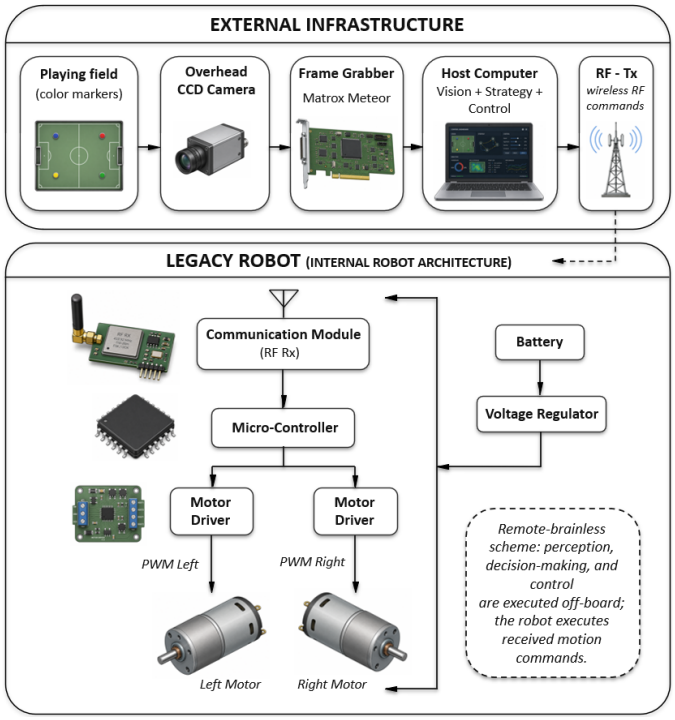


Fig. 2. Original remote-brainless architecture of the legacy FIRA robot. Perception, strategy, and control were executed by the host computer, while the robot only received wireless motion commands.

the mechanical structure was intentionally simple, while perception, strategy, and supervision were executed externally. Consequently, the robot was conceived as part of a larger vision-based soccer system rather than as a fully autonomous mobile robot. Figure 1 shows the original robot together with the legacy competition setup, highlighting the dependence on external infrastructure for perception, strategy, and command generation.

A. Original System Architecture

The original platform followed a remote-brainless architecture. In this configuration, an overhead camera observed the playing field while a host computer performed image processing, robot localization, ball detection, strategy selection, and command generation. Motion commands were then transmitted to the robot through a wireless communication link for execution. As a result, the robot functioned primarily as a mobile actuator, with most of the sensing and decision-making performed off-board.

This architecture was well suited to early FIRA/MiroSot competitions, where the limited size of the robots constrained onboard computation and sensing. However, it also made the platform highly dependent on external infrastructure, including the overhead camera, host computer, frame grabber, RF communication system, and legacy software environment. Figure 2 illustrates the original centralized architecture.

B. Mechanical and Electronic Structure

Mechanically, the robot consists of a compact differential-drive base with two DC motors, wheels, and a lightweight chassis. Although originally intended for robot soccer, the mechanical platform remains suitable for educational activities involving mobile robotics, motion control, and multi-robot coordination. Consequently, the proposed retrofit preserves the original mechanical structure while replacing only the obsolete electronic and software components.

The original electronic system was primarily responsible for receiving wireless commands and driving the motors. It consisted of a microcontroller-based control board, motor driver circuitry, a power regulation stage, batteries, and an RF communication module. The platform also relied on visual markers mounted on top of the robot so that the external vision system could estimate its position and orientation.

C. Obsolescence and Limitations

Although the original system was appropriate for its competition context, it presents several limitations for current robotics education and research. First, the robot lacks onboard sensing, preventing local perception and interaction with its environment. Second, its dependence on external vision and wireless command transmission makes the platform impractical once the original infrastructure is no longer available. Third, the supporting hardware and software rely on obsolete technologies, including legacy frame grabbers, outdated operating systems, and monolithic host-side applications.

From a modern robotics perspective, these limitations significantly reduce the reuse potential of the platform. The robot cannot directly interface with ROS 2 tools, publish sensor information, or be easily extended with additional perception and control modules. Consequently, the primary challenge is not mechanical redesign but architectural modernization: preserving the existing mobile base while replacing the obsolete electronic, communication, and software layers.

These limitations motivate the retrofitting architecture proposed in this paper. Rather than discarding the existing platform or redesigning it from scratch, the proposed approach preserves the mechanically functional robot while replacing the obsolete electronics, communication interfaces, and software with a modular ROS 2/micro-ROS-enabled embedded architecture.

IV. Proposed Architecture

The proposed architecture transforms the legacy FIRA robot into a modular ROS 2-enabled platform while preserving its differential-drive base. It comprises four layers: mechanical hardware, embedded processing, communication, and the ROS 2 host. Unlike the original centralized system, onboard controllers manage sensing and actuation, while the external computer provides supervision and software integration, as illustrated in Figure 3.

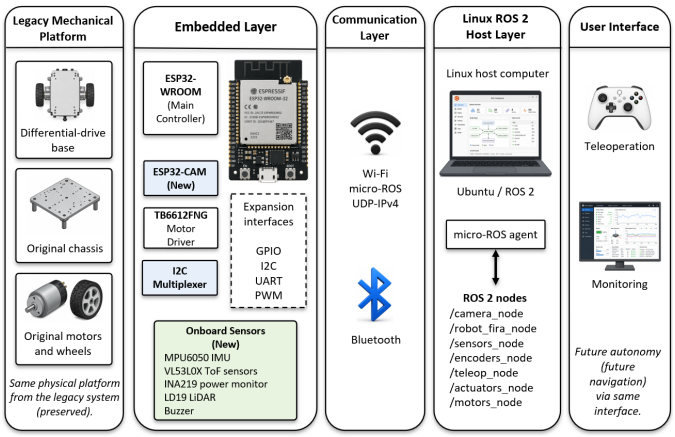


Fig. 3. Proposed ROS 2/micro-ROS retrofitting architecture for the legacy FIRA robot.

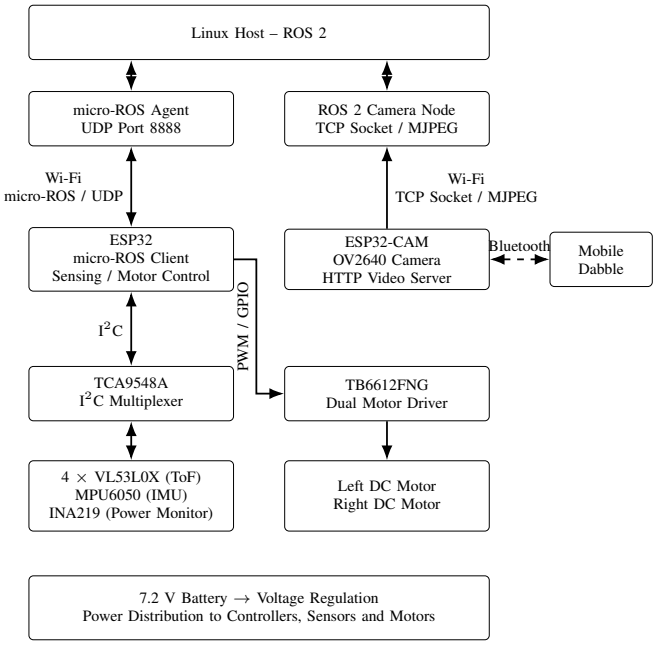


Fig. 4. Hardware interfaces and communication architecture of the retrofitted FIRA robot. The ESP32 communicates through micro-ROS, while the ESP32-CAM provides an independent socket-based video stream to the ROS 2 host.

A. Hardware Layer

The hardware layer preserves the original chassis and motors while replacing obsolete electronics. An ESP32 serves as the primary controller, selected for its integrated Wi-Fi/Bluetooth connectivity, peripheral interfaces, micro-ROS compatibility, and suitability for embedded sensing and actuation. An ESP32-CAM provides dedicated video acquisition, separating camera-related tasks from the main controller.

A dual motor driver independently controls both wheels. An IMU, time-of-flight sensors, and a voltage/current sensor provide onboard measurements, while expansion interfaces accommodate additional peripherals. Modular off-the-shelf components facilitate implementation and future hardware modifications.

B. Embedded Layer

The ESP32 manages sensor acquisition, motor commands, auxiliary actuators, operating modes, and micro-ROS communication. Incoming commands are translated into PWM and direction signals, while sensor measurements are prepared for publication.

The ESP32-CAM handles video streaming and Bluetooth-based manual control. This functional separation assigns camera-related operations and primary sensing and actuation to dedicated embedded modules.

C. Communication Layer

Two operating modes are supported: Bluetooth for direct teleoperation without ROS 2 and Wi-Fi for integration with the ROS 2 environment.

During Wi-Fi operation, the ESP32 exchanges information with a host-based micro-ROS agent through Micro XRCE-DDS [4], [5]. This enables bidirectional sensor and actuator communication through ROS 2 topics. Camera streaming is handled separately by the ESP32-CAM.

D. ROS 2 Host Layer

A Linux-based host executes the micro-ROS agent and custom ROS 2 nodes for motor control, sensor aggregation, encoder processing, actuator management, camera visualization, and teleoperation.

The publish/subscribe architecture separates embedded hardware functions from higher-level applications [3], supporting future navigation and multi-robot coordination.

E. Architectural Transition

The retrofit redistributes responsibilities previously concentrated in the external computer. Onboard controllers now handle sensing, low-level actuation, and communication, while the host provides supervision and software integration.

Rather than eliminating external computation, the proposed framework establishes modular embedded functionality and standardized interfaces, providing a reusable foundation for modernizing legacy robotic platforms.

V. Prototype Implementation

The proposed architecture was implemented as a functional prototype using the original FIRA chassis, motors, and differential-drive mechanism. Modular off-the-shelf electronics replace the obsolete control and communication subsystems, providing a practical implementation of the proposed framework.

A. Hardware Implementation

The prototype integrates an ESP32 as the primary controller, an ESP32-CAM for video acquisition, and a TB6612FNG driver for independent wheel actuation. Onboard sensing includes an MPU6050 IMU, four VL53L0X distance sensors connected through a TCA9548A I^2C multiplexer, and an INA219 power monitor.

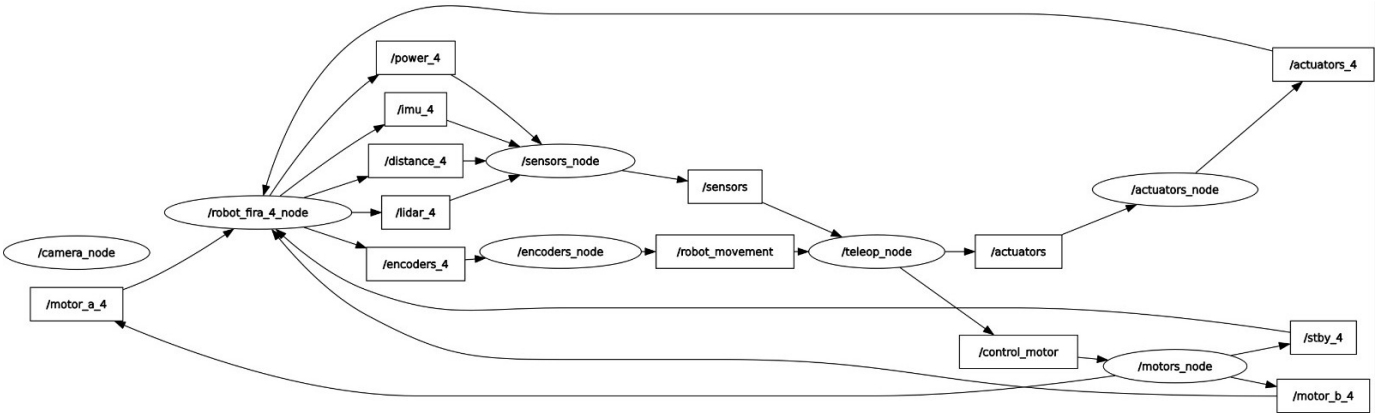


Fig. 5. ROS 2 computation graph of the implemented prototype, illustrating communication among sensing, teleoperation, motor-control, and actuator-management nodes.

Figure 4 details the implemented hardware interfaces and communication paths. The ESP32 controls the motor driver through PWM/GPIO signals and communicates with the ROS 2 host via micro-ROS over Wi-Fi/UDP. Independently, the ESP32-CAM provides video streaming to the host's camera node through a Wi-Fi TCP socket. Bluetooth enables manual control through the Dabble application. A regulated power subsystem supplies the embedded modules, sensors, and motors.

The prototype uses off-the-shelf components and provides expansion interfaces. A custom PCB and more compact hardware integration remain future developments.

B. Software and Communication Implementation

The ESP32 firmware implements sensor acquisition, motor control, operating-mode management, and a micro-ROS client. The ESP32-CAM handles video streaming and Bluetooth-based manual control.

Bluetooth provides direct teleoperation without requiring ROS 2. Alternatively, Wi-Fi connects the ESP32 micro-ROS client to a host-based micro-ROS agent, enabling sensor publication and actuator commands through ROS 2 topics.

The host executes custom ROS 2 nodes for motor control, sensor aggregation, encoder processing, actuator management, camera visualization, and teleoperation. Their implemented organization is illustrated in Figure 5.

VI. Architectural Analysis

Since the original hardware and software stack is no longer operational, a direct experimental comparison is not feasible. Accordingly, this section provides a qualitative architectural assessment of the implemented retrofit, focusing on functional capabilities and software organization rather than quantitative performance.

A. Legacy-to-ROS 2 Transformation

The principal architectural change is the redistribution of functionality between the robot and its host. Unlike the original command-driven platform, the retrofitted robot incorporates onboard sensing and low-level actuation while retaining the host for supervision and software integration. Table I summarizes these architectural differences without implying measured performance improvements.

TABLE I
Architectural comparison between the legacy and retrofitted systems

Feature	Legacy system	Retrofitted system
System paradigm	Remote-brainless	Embedded ROS 2-enabled platform
Computation	External host	ESP32 and ROS 2 host
Communication	RF command link	Bluetooth, Wi-Fi, micro-ROS
Vision	Overhead CCD camera	Onboard ESP32-CAM stream
Software structure	Legacy host program	Modular ROS 2 nodes
Onboard sensing	Absent or limited	IMU, ToF, power monitoring
Data exchange	Primarily command reception	Bidirectional ROS 2 topics
Expandability	Custom legacy electronics	GPIO, I2C, UART, PWM interfaces
Development workflow	Platform-specific	ROS 2/micro-ROS ecosystem

B. Enabled Capabilities

The implemented prototype incorporates onboard sensing, power monitoring, camera streaming, teleoperation, and bidirectional ROS 2 communication. Its modular interfaces also accommodate additional hardware and software components, although autonomous navigation and other advanced functions remain future extensions.

These observations establish the functional scope of the retrofit, not improvements in latency, sensing accuracy, or motion performance. Supplementary implementation evidence, including hardware photographs, the ROS 2 computation graph, camera streaming, and Bluetooth/Wi-Fi demonstrations, is available in the project repository.

VII. Discussion

Unlike approaches that develop new robotic platforms, this work addresses the modernization of mechanically functional but electronically obsolete robots. Its contribution lies in preserving existing hardware while introducing modular embedded processing, onboard sensing, wireless communication, and ROS 2/micro-ROS integration.

The proposed framework illustrates how a centralized, command-driven robot can be reorganized into a distributed architecture without mechanical redesign. Beyond the implemented prototype, this approach provides a reusable foundation for educational applications and future research extensions. Its contribution is architectural rather than a demonstration of improved robotic performance.

VIII. Limitations and Future Work

The proposed framework has three main limitations. First, the original system is no longer operational, preventing direct experimental comparison. Second, the implementation remains a modular prototype rather than a fully integrated hardware platform. Third, systematic evaluations of communication latency, power consumption, sensing accuracy, motor response, and operational robustness have not been conducted. Accordingly, this work establishes an architectural implementation without claiming quantitative performance improvements.

Future work includes developing a dedicated PCB, improving hardware integration, incorporating LiDAR, and extending the ROS 2 software toward navigation and multi-robot coordination. Quantitative evaluation of communication, sensing, energy consumption, and locomotion will also be considered.

Supplementary materials, including source code, firmware, hardware documentation, prototype photographs, ROS 2 computation graphs, and Bluetooth/Wi-Fi demonstrations, are available in the project repository.[1]

IX. Conclusion

This paper presented and implemented a modular retrofitting architecture that transforms a legacy FIRA differential-drive robot into a ROS 2/micro-ROS-enabled embedded platform while preserving its original mechanical structure. The approach replaces obsolete electronics and communication systems with onboard sensing, motor control, camera streaming, wireless connectivity, and standardized software interfaces.

The principal contribution is the redistribution of functionality between embedded controllers and the ROS 2 host. The ESP32 manages sensing and actuation through micro-ROS, while the ESP32-CAM independently provides socket-based video streaming. This separation establishes a modular architecture that accommodates hardware modifications and additional software components without requiring mechanical redesign.

The functional prototype demonstrates the practical implementation of this architectural transition and provides a documented foundation for reusing legacy robotic hardware in education and research. Although quantitative performance evaluation remains future work, the proposed framework establishes an extensible basis for subsequent hardware integration, autonomous navigation, and multi-robot applications.

References

[1] K.-H. Kim, K.-W. Ko, J.-G. Kim, S.-H. Lee, and H.-S. Cho, "The development of a micro robot system for robot soccer game," in *Proceedings of the 1997 IEEE International Conference on Robotics and Automation*, Albuquerque, NM, USA, 1997, pp. 644–649.

[2] N. G. Bizdoaca, H. Hamdan, D. Coman, M. Hamdan, and K. A. Mutib, "Biomimetic control architecture for robotic cooperative tasks," *Artificial Life and Robotics*, vol. 15, no. 4, pp. 403–407, 2010.

[3] Y. Maruyama, S. Kato, and T. Azumi, "Exploring the performance of ROS2," in *Proceedings of the International Conference on Embedded Software*, ser. EMSOFT '16. Pittsburgh, PA, USA: Association for Computing Machinery, 2016.

[4] micro-ROS, "micro-ROS: ROS 2 for microcontrollers," https://micro.vulcanexus.org/, 2026, accessed: 2026-07-04.

[5] eProsima, *eProsima Micro XRCE-DDS Documentation*, https://micro-xrce-dds.docs.eprosima.com/en/latest/, 2026, accessed: 2026-07-04.

[6] E. Jones, D. Adra, and M. S. Miah, "MAFOSS: Multi-agent framework using open-source software," in *2019 7th International Conference on Mechatronics Engineering (ICOM)*, 2019, pp. 1–6.

[7] N. Linus, C. Fritze, K. Bartsch, and E. Rueckert, "ROMR: A ROS-based open-source mobile robot," *HardwareX*, vol. 14, p. e00426, 2023.

[8] W. Jo, J. Kim, R. Wang, J. Pan, R. K. Senthilkumaran, and B.-C. Min, "SMARTmBOT: A ROS2-based low-cost and open-source mobile robot platform," 2022.

[9] M. H. S. Bomfim, R. A. Gontijo, A. Q. Bracarense, and E. J. Lima, "Overhauling of an ASEA robot IR6 with open architecture," in *2012 IEEE 11th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2012.

[10] R. Samson, K. Goh, K. G. Dobaria, A. Gkanatsios, and H. Yu, "Retrofitting legacy AGV with low-cost LiDAR and industry 4.0 technologies for autonomous navigation," *Proceedings of the IEEE International Conference on Automation and Computing (ICAC)*, 2025.

[1] Project repository: https://github.com/LefferTrochez/Differential-Robot-FIRA.git